E-commerce Sales Analysis using Apache Hive in Hadoop Environment

1. Introduction

In the contemporary digital era, Big Data plays a pivotal role in the operational and strategic decision-making processes of modern businesses. E-commerce platforms generate massive volumes of structured, semi-structured, and unstructured data daily, encompassing user transactions, product catalog interactions, and geographical sales metrics. Traditional relational database management systems (RDBMS) often struggle to process these vast datasets efficiently.

To address these challenges, the Hadoop ecosystem provides a robust, distributed computing framework. At its core, the Hadoop Distributed File System (HDFS) enables the fault-tolerant storage of massive datasets across clusters of commodity hardware. Built on top of Hadoop, Apache Hive acts as a data warehouse infrastructure that facilitates reading, writing, and managing large datasets residing in distributed storage using SQL. Hive projects structure onto this data and allows users to query it using a SQL-like language called HiveQL (HQL). This abstraction eliminates the need to write complex Java MapReduce programs, making large-scale e-commerce data analysis accessible, efficient, and highly scalable.

2. Objectives

The primary objectives of this analytical project are as follows:

- **To store and process large-scale sales data** securely and efficiently utilizing the Hadoop Distributed File System (HDFS).
- **To perform complex data analysis** using Apache Hive by executing MapReduce jobs via high-level HiveQL queries.
- **To derive meaningful business insights** from raw e-commerce sales data, including revenue generation patterns, category performance, and geographic sales distribution.

3. Tools and Technologies Used

- **Hadoop (HDFS):** The underlying storage layer utilized to store the raw e-commerce data file. It provides high throughput access to application data and is suitable for applications with large datasets.
- **Apache Hive:** The data warehousing component used to map the raw text data into a structured table format and query it using SQL-like syntax. Hive internally translates these queries into MapReduce jobs.
- **Cloudera Quickstart VM:** A single-node Hadoop cluster deployment running on a virtual machine. It provides a comprehensive, pre-configured environment containing all essential Hadoop ecosystem components (HDFS, YARN, Hive, etc.) for development and testing.
- **Linux Terminal (Bash):** The command-line interface used to interact with the local file system, execute HDFS shell commands, and launch the Hive CLI interface.

4. Dataset Description

The dataset used for this analysis represents daily transaction records of an e-commerce platform. It is stored locally as a comma-separated text file named sales.txt.

The structured schema maps to the following six columns:

1. **order_id (INT):** A unique identifier for each transaction.
2. **product (STRING):** The name of the item purchased.
3. **category (STRING):** The overarching retail category of the product.
4. **price (INT):** The unit price of the product.
5. **quantity (INT):** The number of units purchased in the transaction.
6. **city (STRING):** The geographical location where the order was placed.

Sample Dataset Entries:

1,Mobile,Electronics,15000,2,Delhi

2,Laptop,Electronics,55000,1,Mumbai

3,Shoes,Fashion,2000,3,Delhi

4,Tshirt,Fashion,800,5,Pune

5,Headphones,Electronics,1500,4,Delhi

6,Watch,Accessories,3000,2,Mumbai

7,Bag,Accessories,1200,3,Pune

5. Step-by-Step Implementation

a) File Creation and Data Entry

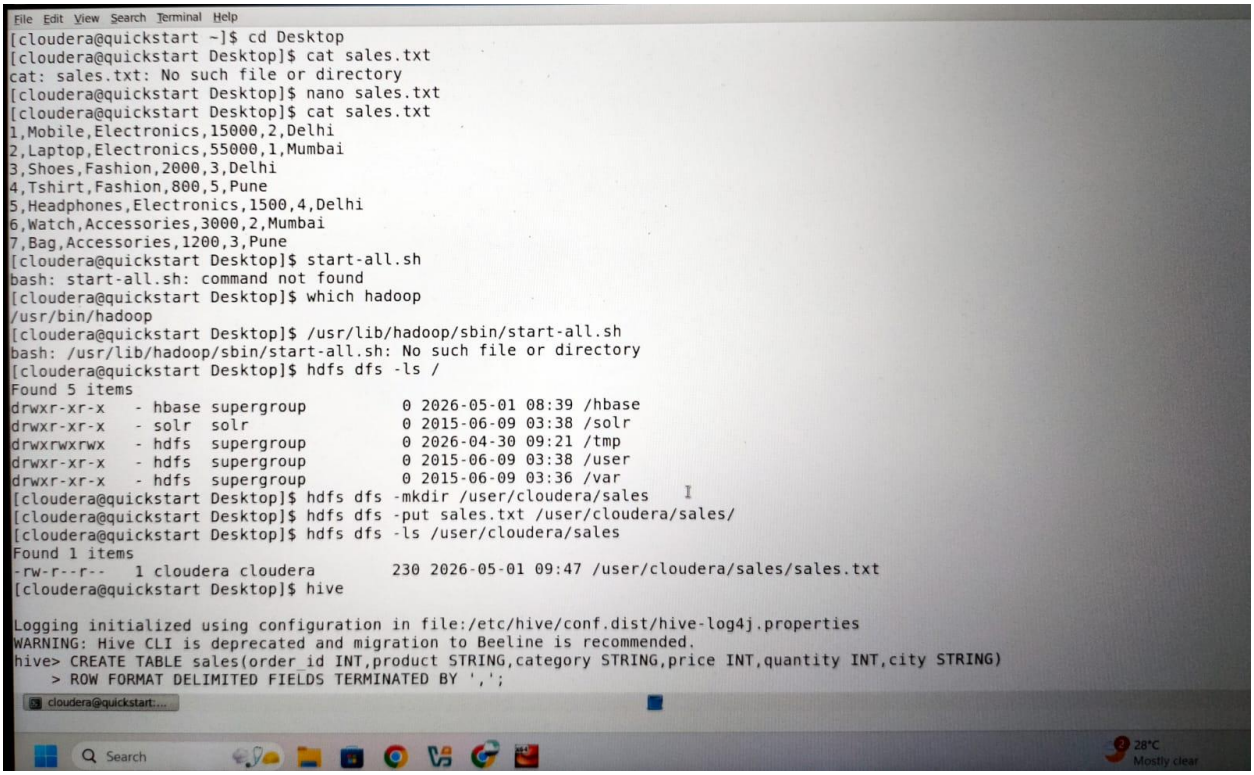
The initial step involved creating the raw data file within the local Linux environment.

- **Command:** nano sales.txt
  - *Purpose:* Opens the nano text editor to input the comma-separated sales records.
- **Command:** cat sales.txt
  - *Purpose:* Displays the contents of the newly created file in the terminal to verify data integrity before uploading it to Hadoop.

b) Hadoop Commands

The local file was then transferred to the Hadoop Distributed File System.

- **Command:** hdfs dfs -mkdir /user/cloudera/sales
  - *Purpose:* Creates a dedicated directory structure within HDFS to store the sales data.
- **Command:** hdfs dfs -put sales.txt /user/cloudera/sales/
  - *Purpose:* Copies the sales.txt file from the local desktop into the designated HDFS directory.
- **Command:** hdfs dfs -ls /user/cloudera/sales
  - *Purpose:* Lists the contents of the HDFS directory to verify that the file was successfully uploaded.



```
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ cd Desktop
[cloudera@quickstart Desktop]$ cat sales.txt
cat: sales.txt: No such file or directory
[cloudera@quickstart Desktop]$ nano sales.txt
[cloudera@quickstart Desktop]$ cat sales.txt
1,Mobile,Electronics,15000,2,Delhi
2,Laptop,Electronics,55000,1,Mumbai
3,Shoes,Fashion,2000,3,Delhi
4,Tshirt,Fashion,800,5,Pune
5,Headphones,Electronics,1500,4,Delhi
6,Watch,Accessories,3000,2,Mumbai
7,Bag,Accessories,1200,3,Pune
[cloudera@quickstart Desktop]$ start-all.sh
bash: start-all.sh: command not found
[cloudera@quickstart Desktop]$ which hadoop
/usr/bin/hadoop
[cloudera@quickstart Desktop]$ /usr/lib/hadoop/sbin/start-all.sh
bash: /usr/lib/hadoop/sbin/start-all.sh: No such file or directory
[cloudera@quickstart Desktop]$ hdfs dfs -ls /
Found 5 items
drwxr-xr-x   - hbase supergroup          0 2026-05-01 08:39 /hbase
drwxr-xr-x   - solr  solr                0 2015-06-09 03:38 /solr
drwxrwxrwx   - hdfs  supergroup          0 2026-04-30 09:21 /tmp
drwxr-xr-x   - hdfs  supergroup          0 2015-06-09 03:38 /user
drwxr-xr-x   - hdfs  supergroup          0 2015-06-09 03:36 /var
[cloudera@quickstart Desktop]$ hdfs dfs -mkdir /user/cloudera/sales
[cloudera@quickstart Desktop]$ hdfs dfs -put sales.txt /user/cloudera/sales/
[cloudera@quickstart Desktop]$ hdfs dfs -ls /user/cloudera/sales
Found 1 items
-rw-r--r--   1 cloudera cloudera        230 2026-05-01 09:47 /user/cloudera/sales/sales.txt
[cloudera@quickstart Desktop]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive> CREATE TABLE sales(order_id INT,product STRING,category STRING,price INT,quantity INT,city STRING)
    > ROW FORMAT DELIMITED FIELDS TERMINATED BY ',';
```

c) Hive Table Creation

With the data residing in HDFS, the Hive CLI was launched to apply a structural schema.

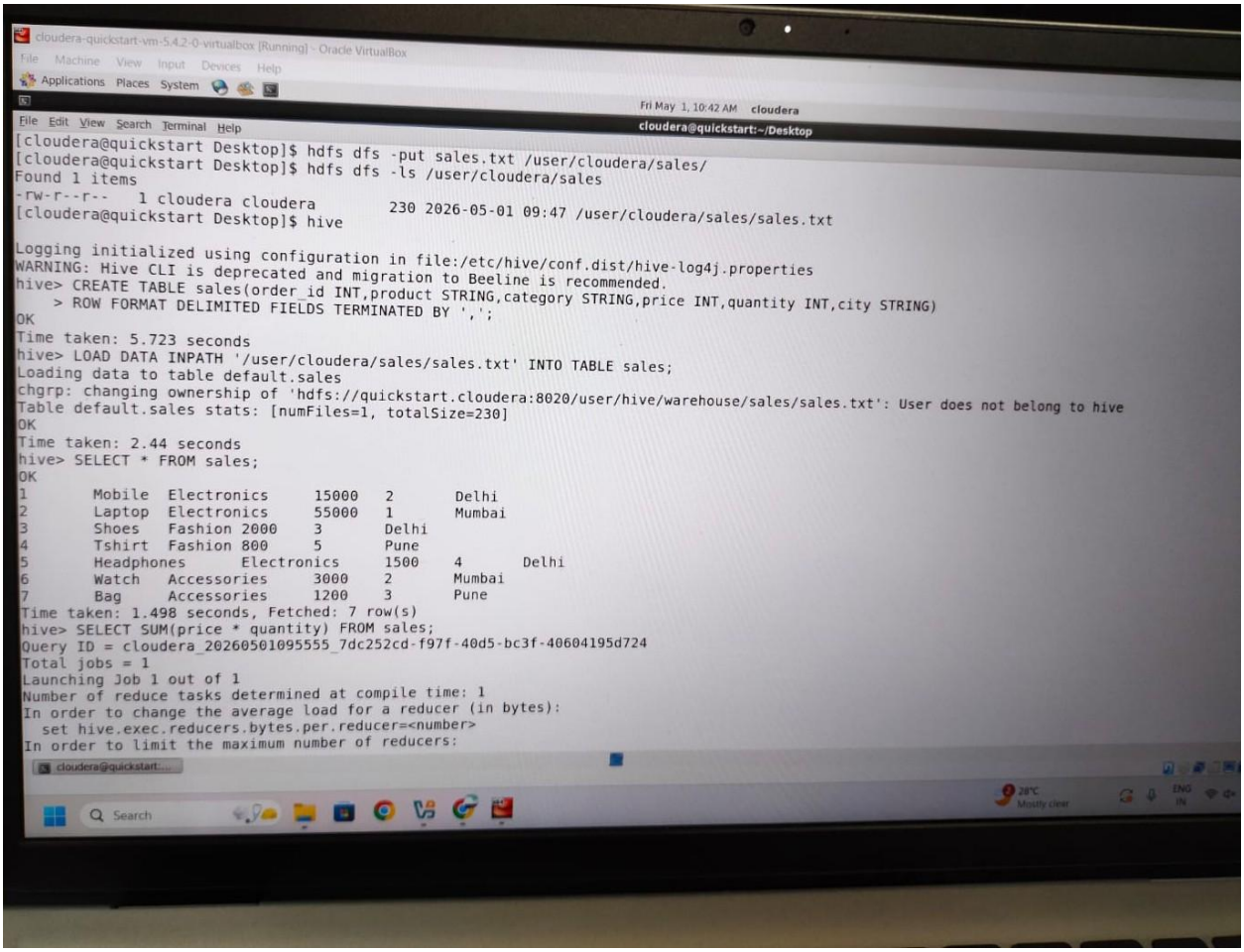
- **Command:**
  CREATE TABLE sales(order_id INT, product STRING, category STRING, price INT, quantity INT, city STRING)
  ROW FORMAT DELIMITED FIELDS TERMINATED BY ',';

  - *Purpose:* This DDL (Data Definition Language) command creates a table named sales and explicitly defines the data types for each column. The ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' clause is crucial; it instructs Hive's SerDe (Serializer/Deserializer) to parse the incoming text file by treating commas as column separators.

d) Loading Data into Hive

- **Command:**
  LOAD DATA INPATH '/user/cloudera/sales/sales.txt' INTO TABLE sales;

  - *Purpose:* This command moves the data file from its current HDFS path into the Hive warehouse directory associated with the sales table. Hive does not perform data transformation on load; it simply moves the file and applies the schema upon reading (schema-on-read).



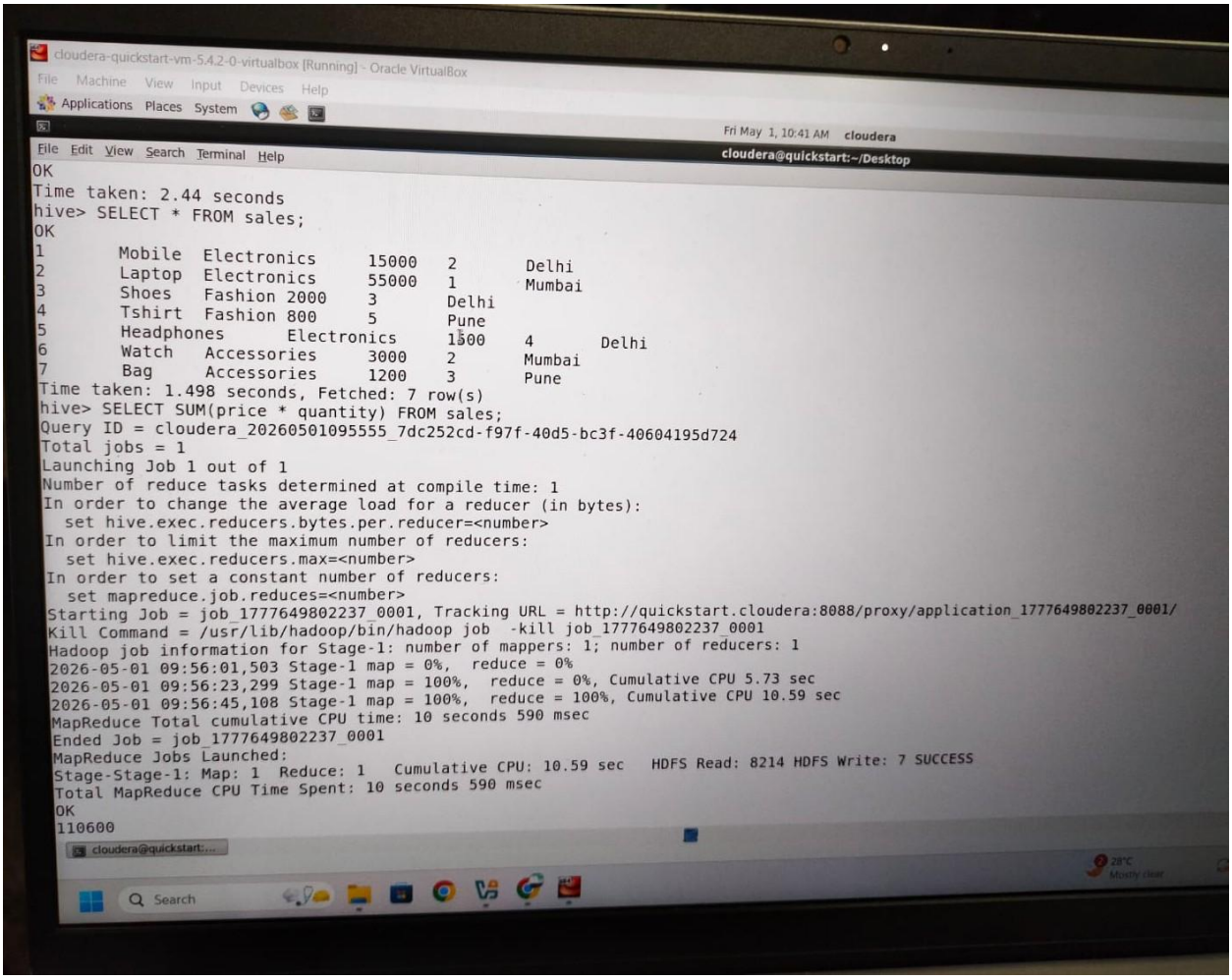
```
[cloudera@quickstart Desktop]$ hdfs dfs -put sales.txt /user/cloudera/sales/
[cloudera@quickstart Desktop]$ hdfs dfs -ls /user/cloudera/sales
Found 1 items
-rw-r--r--   1 cloudera cloudera        230 2026-05-01 09:47 /user/cloudera/sales/sales.txt
[cloudera@quickstart Desktop]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive> CREATE TABLE sales(order_id INT,product STRING,category STRING,price INT,quantity INT,city STRING)
    > ROW FORMAT DELIMITED FIELDS TERMINATED BY ',';
OK
Time taken: 5.723 seconds
hive> LOAD DATA INPATH '/user/cloudera/sales/sales.txt' INTO TABLE sales;
Loading data to table default.sales
chgrp: changing ownership of 'hdfs://quickstart.cloudera:8020/user/hive/warehouse/sales/sales.txt': User does not belong to hive
Table default.sales stats: [numFiles=1, totalSize=230]
OK
Time taken: 2.44 seconds
hive> SELECT * FROM sales;
OK
1       Mobile  Electronics     15000   2       Delhi
2       Laptop  Electronics     55000   1       Mumbai
3       Shoes   Fashion 2000    3       Delhi
4       Tshirt  Fashion 800     5       Pune
5       Headphones      Electronics     1500    4       Delhi
6       Watch   Accessories     3000    2       Mumbai
7       Bag     Accessories     1200    3       Pune
Time taken: 1.498 seconds, Fetched: 7 row(s)
hive> SELECT SUM(price * quantity) FROM sales;
Query ID = cloudera_20260501095555_7dc252cd-f97f-40d5-bc3f-40604195d724
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
```

e) Data Analysis using Hive Queries

Once the table was populated, analytical queries were executed. *Note: Aggregation queries in Hive automatically trigger underlying MapReduce jobs to process the data across the cluster.*

1. Display all records

- **Command:** SELECT * FROM sales;
- **Purpose:** Retrieves all rows and columns from the table to verify the data was parsed correctly.
- **Output:** A tabular display of all 7 transaction records.



```
OK
Time taken: 2.44 seconds
hive> SELECT * FROM sales;
OK
1       Mobile  Electronics     15000   2       Delhi
2       Laptop  Electronics     55000   1       Mumbai
3       Shoes   Fashion 2000    3       Delhi
4       Tshirt  Fashion 800     5       Pune
5       Headphones      Electronics     1500    4       Delhi
6       Watch   Accessories     3000    2       Mumbai
7       Bag     Accessories     1200    3       Pune
Time taken: 1.498 seconds, Fetched: 7 row(s)
hive> SELECT SUM(price * quantity) FROM sales;
Query ID = cloudera_20260501095555_7dc252cd-f97f-40d5-bc3f-40604195d724
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1777649802237_0001, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1777649802237_0001/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0001
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-05-01 09:56:01,503 Stage-1 map = 0%,  reduce = 0%
2026-05-01 09:56:23,299 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 5.73 sec
2026-05-01 09:56:45,108 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 10.59 sec
MapReduce Total cumulative CPU time: 10 seconds 590 msec
Ended Job = job_1777649802237_0001
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 10.59 sec   HDFS Read: 8214 HDFS Write: 7 SUCCESS
Total MapReduce CPU Time Spent: 10 seconds 590 msec
OK
110600
```

2. Calculate total revenue

- **Command:** SELECT SUM(price * quantity) FROM sales;
- **Purpose:** Computes the gross revenue generated across all sales by multiplying the unit price by the quantity for each row, then summing the results.
- **Output:** 110600

```
File Edit View Search Terminal Help
hive> SELECT SUM(price * quantity) FROM sales;
Query ID = cloudera_20260501095555_7dc252cd-f97f-40d5-bc3f-40604195d724
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
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In order to set a constant number of reducers:
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Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0001
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-05-01 09:56:01,503 Stage-1 map = 0%,  reduce = 0%
2026-05-01 09:56:23,299 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 5.73 sec
2026-05-01 09:56:45,108 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 10.59 sec
MapReduce Total cumulative CPU time: 10 seconds 590 msec
Ended Job = job_1777649802237_0001
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 10.59 sec   HDFS Read: 8214 HDFS Write: 7 SUCCESS
Total MapReduce CPU Time Spent: 10 seconds 590 msec
OK
110600
Time taken: 79.331 seconds, Fetched: 1 row(s)
hive> SELECT product,SUM(quantity)
    > FROM sales
    > GROUP BY product
    > ORDER BY SUM(quantity) DESC;
FAILED: SemanticException [Error 10004]: Line 4:13 Invalid table alias or column reference 'quantity': (possible column names are: product, _c1)
hive> SELECT product, SUM(quantity)
    > FROM sales
    > GROUP BY product
    > ORDER BY SUM(quantity) DESC;
FAILED: SemanticException [Error 10004]: Line 4:13 Invalid table alias or column reference 'quantity': (possible column names are: product, _c1)
hive> DESCRIBE sales;
cloudera@quickstart:...
```
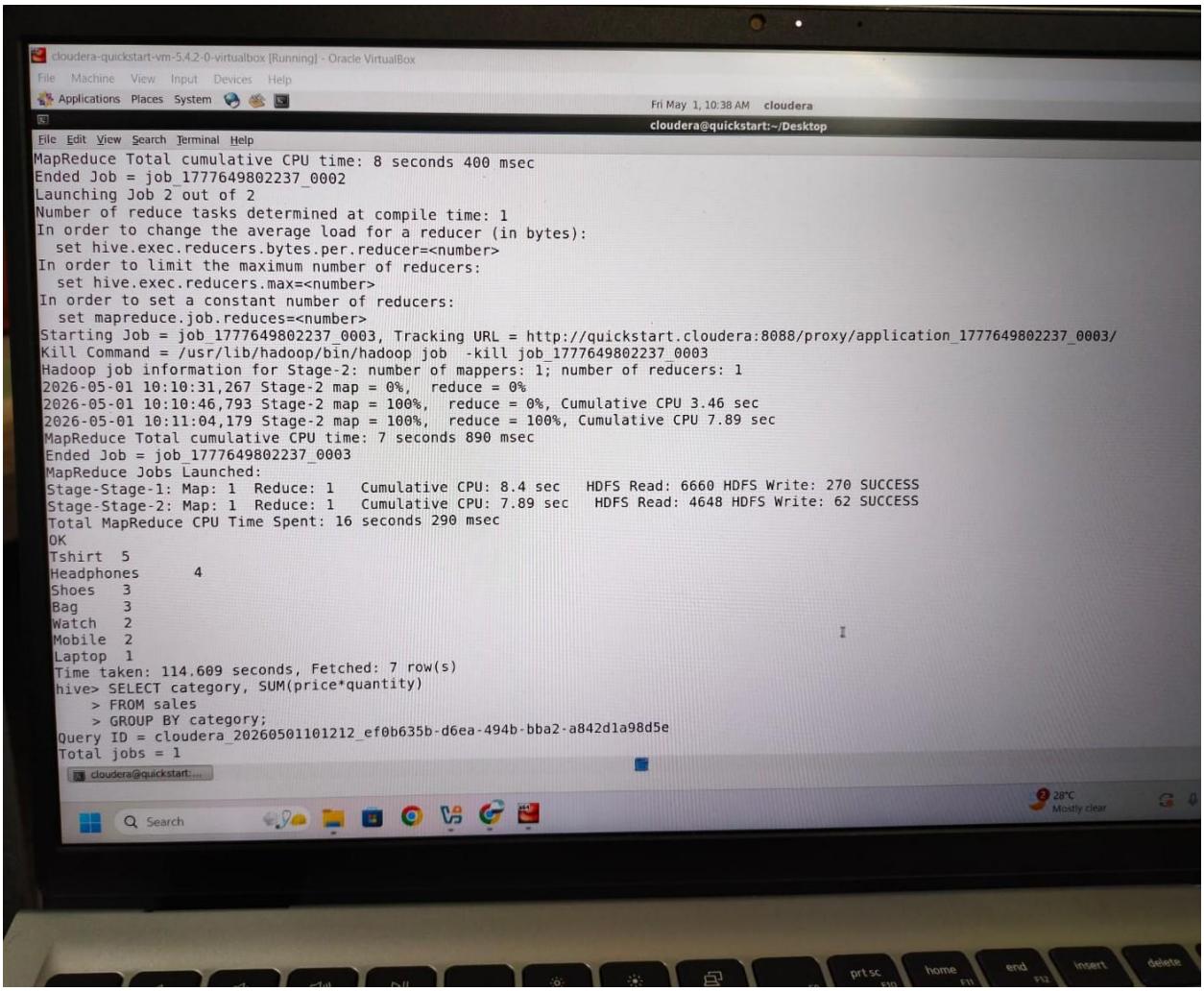

3. Product-wise total quantity

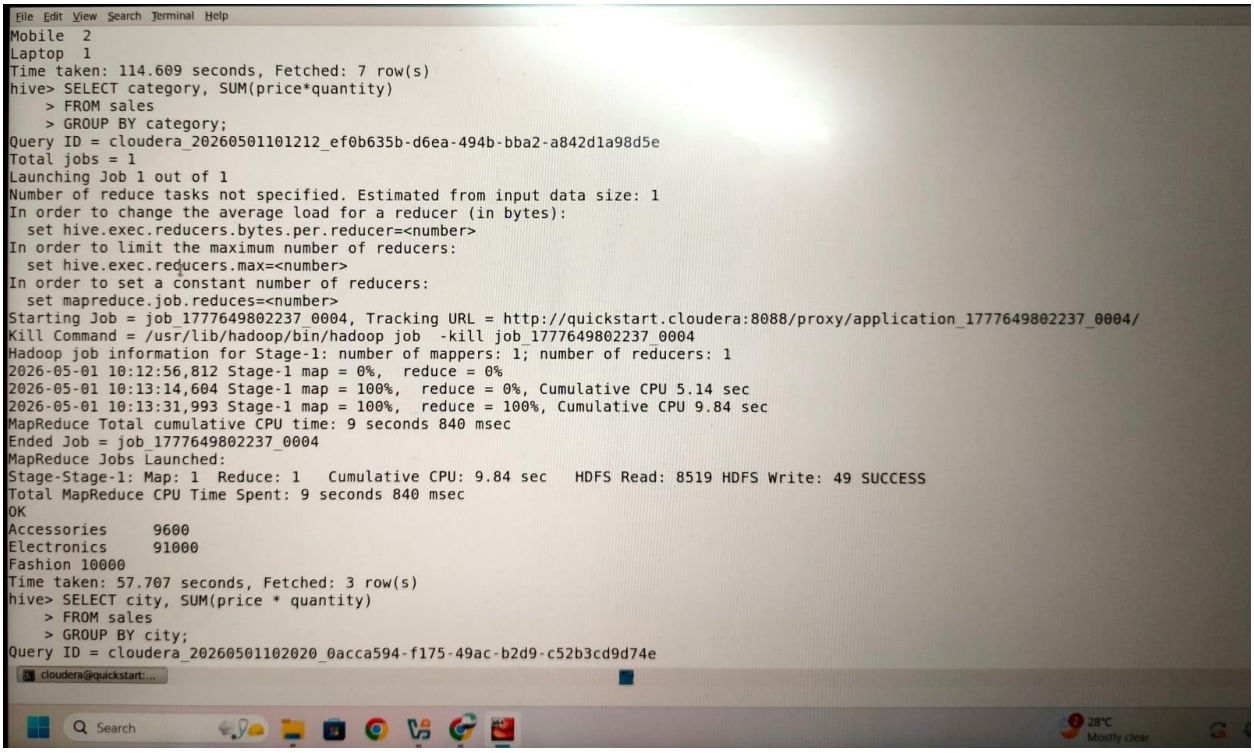
- **Command:** SELECT product, SUM(quantity) AS total_quantity FROM sales GROUP BY product ORDER BY total_quantity DESC;
- **Purpose:** Groups the data by distinct products and calculates the total units sold for each, sorting the output in descending order to identify the most popular items.
- **Output:** Tshirt (5), Headphones (4), Shoes (3), Bag (3), Watch (2), Mobile (2), Laptop (1).

```
File Edit View Search Terminal Help
hive> SELECT product, SUM(quantity) AS total_quantity
    > FROM sales
    > GROUP BY product
    > ORDER BY total_quantity DESC;
Query ID = cloudera_20260501100909_43d34222-52fb-4b1f-be96-9ed4a6810221
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1777649802237_0002, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1777649802237_0002/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0002
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-05-01 10:09:33,339 Stage-1 map = 0%,  reduce = 0%
2026-05-01 10:09:51,240 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 3.88 sec
2026-05-01 10:10:09,768 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 8.4 sec
MapReduce Total cumulative CPU time: 8 seconds 400 msec
Ended Job = job_1777649802237_0002
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1777649802237_0003, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1777649802237_0003/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0003
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2026-05-01 10:10:31,267 Stage-2 map = 0%,  reduce = 0%
2026-05-01 10:10:46,793 Stage-2 map = 100%,  reduce = 0%, Cumulative CPU 3.46 sec
2026-05-01 10:11:04,179 Stage-2 map = 100%,  reduce = 100%, Cumulative CPU 7.89 sec
cloudera@quickstart:...
```



4. Category-wise revenue

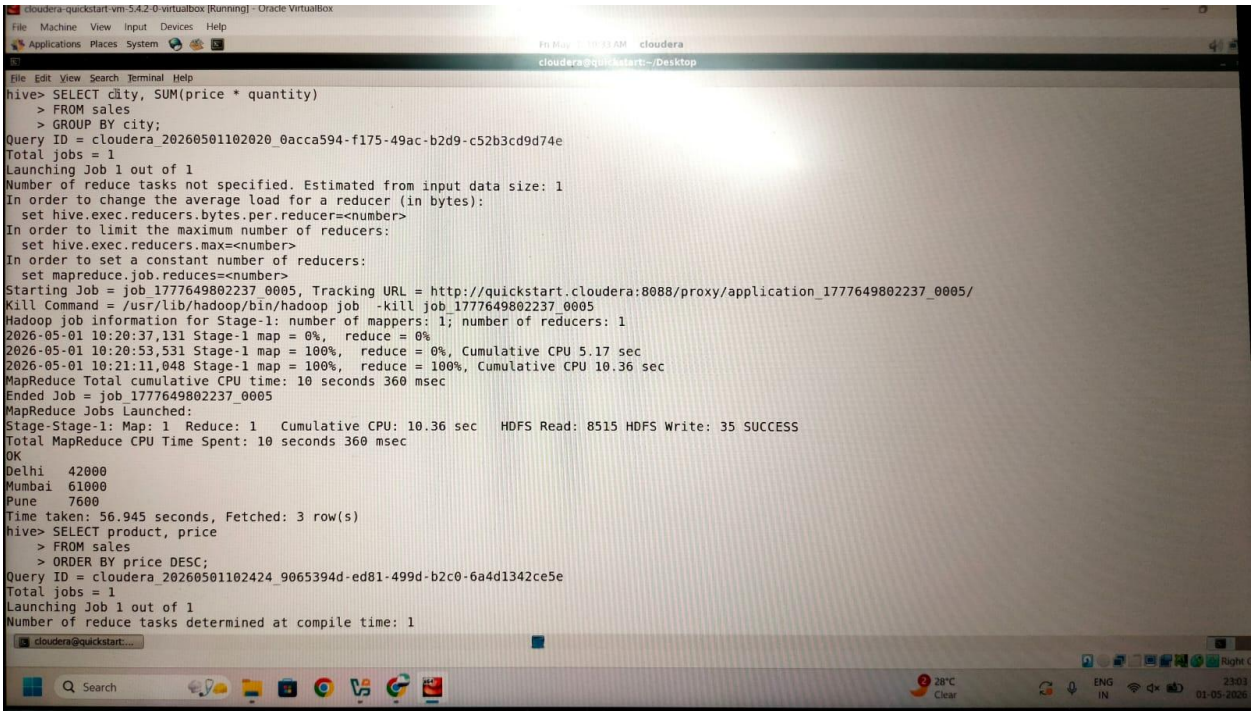
- **Command:** SELECT category, SUM(price*quantity) FROM sales GROUP BY category;
- **Purpose:** Aggregates total sales revenue based on the product category to determine which business vertical is the most lucrative.
- **Output:** Accessories (9600), Electronics (91000), Fashion (10000).



```
File Edit View Search Terminal Help
Mobile  2
Laptop  1
Time taken: 114.609 seconds, Fetched: 7 row(s)
hive> SELECT category, SUM(price*quantity)
    > FROM sales
    > GROUP BY category;
Query ID = cloudera_20260501101212_ef0b635b-d6ea-494b-bba2-a842d1a98d5e
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1777649802237_0004, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1777649802237_0004/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0004
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-05-01 10:12:56,812 Stage-1 map = 0%,  reduce = 0%
2026-05-01 10:13:14,604 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 5.14 sec
2026-05-01 10:13:31,993 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 9.84 sec
MapReduce Total cumulative CPU time: 9 seconds 840 msec
Ended Job = job_1777649802237_0004
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 9.84 sec   HDFS Read: 8519 HDFS Write: 49 SUCCESS
Total MapReduce CPU Time Spent: 9 seconds 840 msec
OK
Accessories     9600
Electronics     91000
Fashion 10000
Time taken: 57.707 seconds, Fetched: 3 row(s)
hive> SELECT city, SUM(price * quantity)
    > FROM sales
    > GROUP BY city;
Query ID = cloudera_20260501102020_0acca594-f175-49ac-b2d9-c52b3cd9d74e
```

5. City-wise revenue

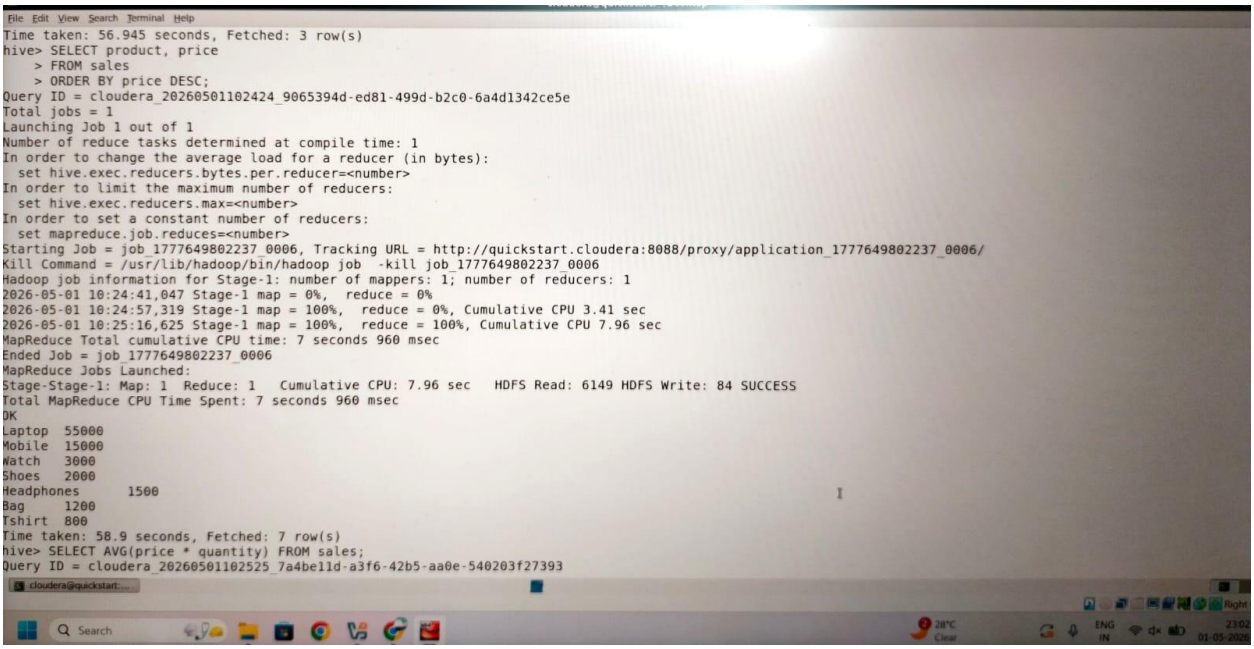
- **Command:** SELECT city, SUM(price * quantity) FROM sales GROUP BY city;
- **Purpose:** Groups transactions by geographical location to analyze regional sales performance.
- **Output:** Delhi (42000), Mumbai (61000), Pune (7600).



```
File Edit View Search Terminal Help
hive> SELECT city, SUM(price * quantity)
    > FROM sales
    > GROUP BY city;
Query ID = cloudera_20260501102020_0acca594-f175-49ac-b2d9-c52b3cd9d74e
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1777649802237_0005, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1777649802237_0005/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0005
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-05-01 10:20:37,131 Stage-1 map = 0%,  reduce = 0%
2026-05-01 10:20:53,531 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 5.17 sec
2026-05-01 10:21:11,048 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 10.36 sec
MapReduce Total cumulative CPU time: 10 seconds 360 msec
Ended Job = job_1777649802237_0005
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 10.36 sec   HDFS Read: 8515 HDFS Write: 35 SUCCESS
Total MapReduce CPU Time Spent: 10 seconds 360 msec
OK
Delhi   42000
Mumbai  61000
Pune    7600
Time taken: 56.945 seconds, Fetched: 3 row(s)
hive> SELECT product, price
    > FROM sales
    > ORDER BY price DESC;
Query ID = cloudera_20260501102424_9065394d-ed81-499d-b2c0-6a4d1342ce5e
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
```

6. Sort products by price

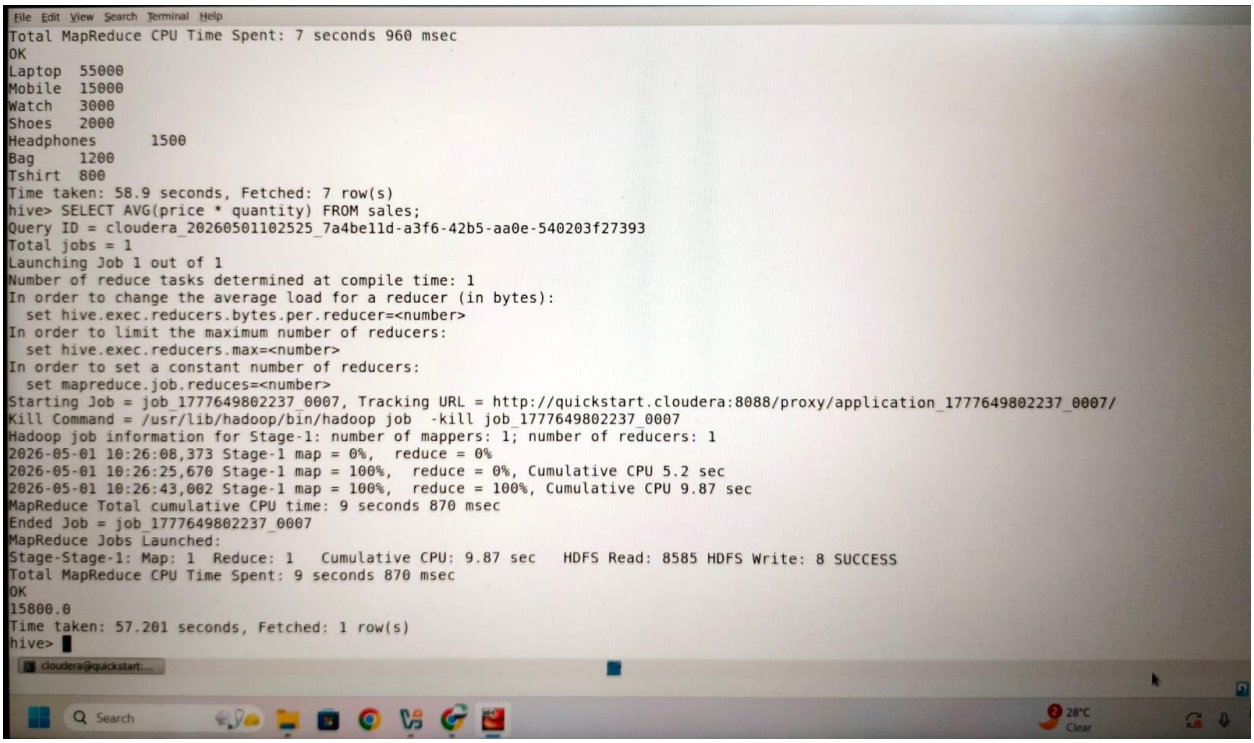
- **Command:** SELECT product, price FROM sales ORDER BY price DESC;
- **Purpose:** Lists all products ordered by their unit price from highest to lowest.
- **Output:** Laptop (55000), Mobile (15000), Watch (3000), Shoes (2000), Headphones (1500), Bag (1200), Tshirt (800).



```
File Edit View Search Terminal Help
Time taken: 56.945 seconds, Fetched: 3 row(s)
hive> SELECT product, price
    > FROM sales
    > ORDER BY price DESC;
Query ID = cloudera_20260501102424_9065394d-ed81-499d-b2c0-6a4d1342ce5e
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1777649802237_0006, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1777649802237_0006/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0006
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-05-01 10:24:41,047 Stage-1 map = 0%,  reduce = 0%
2026-05-01 10:24:57,319 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 3.41 sec
2026-05-01 10:25:16,625 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 7.96 sec
MapReduce Total cumulative CPU time: 7 seconds 960 msec
Ended Job = job_1777649802237_0006
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 7.96 sec   HDFS Read: 6149 HDFS Write: 84 SUCCESS
Total MapReduce CPU Time Spent: 7 seconds 960 msec
OK
Laptop  55000
Mobile  15000
Watch   3000
Shoes   2000
Headphones      1500
Bag     1200
Tshirt  800
Time taken: 58.9 seconds, Fetched: 7 row(s)
hive> SELECT AVG(price * quantity) FROM sales;
Query ID = cloudera_20260501102525_7a4be11d-a3f6-42b5-aa0e-540203f27393
```

7. Average revenue calculation

- **Command:** SELECT AVG(price * quantity) FROM sales;
- **Purpose:** Calculates the Average Order Value (AOV) by finding the mean of the total transaction values.
- **Output:** 15800.0



```
File  Edit  View  Search  Terminal  Help
Total MapReduce CPU Time Spent: 7 seconds 960 msec
OK
Laptop  55000
Mobile  15000
Watch   3000
Shoes   2000
Headphones      1500
Bag     1200
Tshirt  800
Time taken: 58.9 seconds, Fetched: 7 row(s)
hive> SELECT AVG(price * quantity) FROM sales;
Query ID = cloudera_20260501102525_7a4be11d-a3f6-42b5-aa0e-540203f27393
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1777649802237_0007, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1777649802237_0007/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1777649802237_0007
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-05-01 10:26:08,373 Stage-1 map = 0%,  reduce = 0%
2026-05-01 10:26:25,670 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 5.2 sec
2026-05-01 10:26:43,002 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 9.87 sec
MapReduce Total cumulative CPU time: 9 seconds 870 msec
Ended Job = job_1777649802237_0007
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 9.87 sec   HDFS Read: 8585 HDFS Write: 8 SUCCESS
Total MapReduce CPU Time Spent: 9 seconds 870 msec
OK
15800.0
Time taken: 57.201 seconds, Fetched: 1 row(s)
hive>
```

6. Results and Insights

Based on the execution of the HiveQL queries, several critical business insights can be derived from the sample dataset:

- **Top Revenue Driver (Category):** The *Electronics* category vastly outperforms all other sectors, generating a total revenue of ₹91,000, which accounts for the majority of the total gross sales (₹110,600).
- **Geographic Stronghold:** *Mumbai* is the highest-grossing city, bringing in ₹61,000. *Delhi* follows with ₹42,000. Marketing efforts and supply chain logistics should prioritize these metropolitan hubs.
- **Volume vs. Value:** While the *Laptop* generates the highest revenue per unit (₹55,000) and drives the Mumbai market, the highest volume product is the *T-shirt* (5 units sold). This indicates a healthy mix of high-ticket, low-volume sales alongside low-ticket, high-volume sales.
- **Average Order Value:** The business maintains a strong Average Order Value (AOV) of ₹15,800, heavily influenced by the high price of consumer electronics.

7. Conclusion

This project successfully demonstrated the end-to-end process of ingesting, storing, and analyzing e-commerce sales data using the Hadoop ecosystem. By leveraging HDFS for resilient storage and Apache Hive for structural querying, we were able to seamlessly bypass complex Java programming and execute MapReduce jobs using familiar SQL syntax.The analysis yielded actionable business intelligence regarding category profitability, regional performance, and product movement. In real-world enterprise scenarios, this architecture—often scaled across hundreds of nodes—is utilized by massive e-commerce giants (like Amazon or Flipkart) to process petabytes of daily transaction logs, enabling real-time inventory management, dynamic pricing, and targeted customer recommendations.